DSA 210 Final Report

To What Extent Does Film Length Affect Its IMDb Rating?

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Project Summary

Item	Summary
Research question	To what extent does film length affect its IMDb rating?
Main dataset	Public IMDb movie metadata dataset (movie_metadata.csv).
Final EDA sample	4,790 movies after data cleaning.
Main variables	Runtime, IMDb rating, genre, and release year.
ML target	IMDb rating.
Best ML model	Ridge Regression based on MAE.

1. Motivation

Movies are part of everyday life, and IMDb ratings are one of the most common ways people decide what to watch. I chose this topic because movie length is a simple and visible feature, but it is not clear whether it actually affects how people rate a movie. Some longer movies may have more detailed stories and stronger character development, while others may feel slow or tiring. Because of this, the relationship between runtime and rating is interesting to examine with data rather than only personal opinion.

The main goal of this project is to understand whether longer movies tend to receive higher IMDb ratings and whether this relationship remains meaningful when other variables, such as genre and release year, are also considered. The project also gives a practical example of the full data science pipeline: data collection, cleaning, exploratory data analysis, hypothesis testing, machine learning, and interpretation.

2. Research Question

The main research question of this project is:

To what extent does film length affect its IMDb rating?

The supporting questions are:

- Is there a measurable relationship between movie runtime and IMDb rating?
- Do longer movies have higher average IMDb ratings than shorter movies?
- Do genre and release year also matter when interpreting this relationship?
- Can IMDb ratings be predicted using runtime, genre, and release year?

3. Data Source and Data Collection

The project uses a public movie metadata dataset saved as movie_metadata.csv. The dataset contains movie-level information such as movie title, duration, IMDb score, genre, and release year. The original dataset had 5,043 rows. This dataset was suitable for the project because it directly includes the core variables needed for the research question: movie duration and IMDb rating.

The public dataset was downloaded and added to the GitHub repository. I then used Python to load, inspect, clean, and transform the data. Since the project focuses on the relationship between movie length and rating, the main analysis uses runtime and IMDb score, while genre and release year are used as additional explanatory variables. This made the analysis more complete than only comparing duration and rating.

Main variables used in the project:

- duration / runtime: movie length in minutes.
- imdb_score / imdb_rating: IMDb rating of the movie.
- genres / genre: movie genre information, simplified into a primary genre for analysis.
- title_year / release_year: the release year of the movie.

4. Data Cleaning and Preparation

The data cleaning step focused on making the dataset consistent and usable for analysis. I selected the columns related to movie title, duration, IMDb rating, genre, and release year. I converted runtime and IMDb rating into numeric values, removed missing values in key columns, filtered unrealistic runtime and rating values, and removed duplicate movie titles.

After cleaning, 4,790 movies remained in the dataset. The average runtime was about 108.09 minutes, and the average IMDb rating was about 6.41. The median runtime was 104 minutes, which was later used to divide movies into short and long groups for hypothesis testing.

Cleaned Dataset Summary

Variable	Mean	Median	Standard Deviation
Runtime	108.09	104.00	21.93
IMDb rating	6.41	6.50	1.12
Release year	2002.44	2005.00	12.45

5. Exploratory Data Analysis

The exploratory data analysis stage examined the distributions of runtime and IMDb rating, as well as the relationship between them. I used histograms, boxplots, scatter plots, and genre-based comparisons. The runtime histogram and boxplot showed that most movies in the dataset are around 90 to 120 minutes, with some longer outliers. The IMDb rating distribution showed that most movies were concentrated around the middle-to-high range, especially between about 5.5 and 7.5.

The scatter plot between runtime and IMDb rating showed a positive trend. This means that longer movies tended to have higher ratings on average. However, the relationship was not perfect, and there was still a wide spread of ratings for movies with similar runtimes. This suggests that runtime matters, but it is not the only factor affecting ratings.

The genre-based analysis was also important. Some genres appeared to have different typical runtimes and rating distributions. This showed that the runtime-rating relationship may partly depend on genre. For example, genres that usually have longer movies may also receive different ratings for reasons other than runtime itself. Because of this, genre was included later in the machine learning stage.

6. Hypothesis Testing

To support the visual findings with statistical evidence, I conducted correlation tests and group comparison tests. The Pearson correlation between runtime and IMDb rating was 0.3538, and the Spearman correlation was 0.3723. Both results indicate a moderate positive relationship between runtime and rating. The p-values were very small, suggesting that this relationship is statistically significant in the dataset.

I also divided movies into short and long groups using the median runtime of 104 minutes. Short movies had an average IMDb rating of 6.059, while long movies had an average IMDb rating of 6.794. Both the t-test and Mann-Whitney U test gave statistically significant results. This supports the idea that long movies in this dataset tend to receive higher ratings on average.

Finally, I tested whether runtime and rating differed across genres using ANOVA and Kruskal-Wallis tests. These tests were also statistically significant, which means that genre differences should be considered when interpreting the relationship between runtime and rating.

Main Statistical Results

Test / Result	Value	Interpretation
Pearson correlation	$r = 0.3538$, $p < 0.001$	Moderate positive linear relationship.
Spearman correlation	$rho = 0.3723$, $p < 0.001$	Moderate positive rank-based relationship.
Short movie mean rating	6.059	Lower average rating group.
Long movie mean rating	6.794	Higher average rating group.
Genre tests	$p < 0.001$	Runtime and rating differ across genres.

7. Machine Learning Methods

For the machine learning stage, the task was treated as a regression problem because IMDb rating is a numeric variable. The goal was to predict IMDb rating using runtime, release year, and genre. Genre was encoded as a categorical feature, while runtime and release year were handled as numeric features.

I tested several models to compare simple and more flexible approaches:

- Baseline Mean Model
- Linear Regression
- Ridge Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

The baseline model predicts the average IMDb rating and is used as a reference point. A useful model should perform better than this baseline. The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R2 score. MAE is especially easy to interpret because it shows the average prediction error in rating points.

Machine Learning Results

Model	MAE	RMSE	R2
Ridge Regression	0.704	0.924	0.254
Linear Regression	0.705	0.925	0.252
Gradient Boosting	0.717	0.946	0.219
Random Forest	0.719	0.952	0.209
Decision Tree	0.776	1.017	0.096
Baseline Mean	0.825	1.071	-0.002

Ridge Regression performed best based on MAE, with an average error of about 0.704 rating points. This means that the model predictions were usually within about 0.7 IMDb rating points of the actual rating. The model performed better than the baseline, so runtime, genre, and release year do contain some useful predictive signal. However, the R2 score was about 0.254, which means that these variables explain only part of the variation in IMDb ratings.

The tree-based models did not perform better than Ridge Regression. This suggests that, for the selected variables, a simpler linear model with regularization was enough. It also shows that adding more complex algorithms does not automatically guarantee better performance if the input features are limited.

8. Main Findings

- There is a moderate positive relationship between movie runtime and IMDb rating.
- Longer movies had higher average ratings than shorter movies in the cleaned dataset.
- Genre matters and may partly explain differences in runtime and rating.
- Machine learning models performed better than the baseline, but the prediction power was limited.
- Ridge Regression was the best model based on MAE, but the R2 score showed that runtime, genre, and year alone cannot fully explain IMDb ratings.

Overall, the analysis suggests that film length is related to IMDb rating, but it should not be treated as the only or main reason behind ratings. IMDb ratings are subjective and are likely influenced by many other factors, including story quality, acting, director reputation, popularity, number of votes, budget, and audience expectations.

9. Limitations and Future Work

The main limitation of this project is that the dataset only includes a limited number of features. Runtime, genre, and release year are useful, but they cannot capture the full complexity of movie ratings. IMDb ratings are affected by many qualitative and external factors that are not fully represented in this dataset.

Another limitation is that IMDb ratings are subjective. A movie with a high rating may be popular because of cultural impact, fan base, marketing, or historical importance. Also, the number of votes matters because a rating based on many votes may be more stable than a rating based on fewer votes.

In future work, the project could be improved by adding variables such as number of votes, budget, gross revenue, director information, actor information, critic scores, production country, language, and awards. It would also be useful to compare results across genres separately instead of treating all movies together.

10. Ethical Considerations and AI Use Disclosure

This project uses a public movie dataset and does not include private personal data. The analysis is based on movie metadata and public rating information. Since ratings are subjective, the results should not be interpreted as proving that longer movies are objectively better. The findings only describe patterns in the selected dataset.

I used AI tools for support in organizing the project structure, improving wording, debugging code, and planning the analysis steps. The dataset selection, code execution, interpretation of outputs, GitHub submission, and final project decisions were completed by me. AI assistance was used as a support tool, not as a replacement for my own work.

References

- Kaggle. IMDb 5000 Movie Dataset / movie_metadata.csv.
- IMDb. Public movie rating and metadata information represented in the dataset.
- Python libraries used: pandas, NumPy, matplotlib, SciPy, and scikit-learn.